

Adam Bawatneh

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Research Interests

Social reasoning and Theory of Mind in LLMs; structured belief modeling and evaluation.
Natural language processing, language model evaluation, multimodal systems, and responsible AI.

Education

University of Central Florida	Orlando, Florida
Ph.D. in Computer Science	Fall 2025 – Present; GPA: 4.00/4.00
University of Central Florida	Orlando, Florida
B.S. in Computer Science	Spring 2025

Selected Publication

Bawatneh, A., et al. “OmniToM: Benchmarking Theory of Mind in LLMs via Explicit Belief Modeling.”
NeurIPS 2026, accepted, Evaluations & Datasets Track. Project | Benchmark

Research Experience

Graduate Research Assistant, University of Central Florida	May 2026 – Present
Conduct AI research and contribute to government proposal development, including IARPA-focused proposals.	
Undergraduate Teaching Assistant, University of Central Florida	May 2023 – Spring 2025
Led labs and mentored students in C, Python, and Java courses serving more than 1,500 students.	

Selected Research Projects

OmniToM: Social Reasoning Evaluation for LLMs	Oct 2025 – Present
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- Co-developed a benchmark with 895 stories, 22,343 belief propositions, and 156,401 labels; coordinated annotation quality control across 11 annotators and about 1.1K person-hours.
- Built structured extraction and labeling prompts; evaluated API and open-weight models across Azure OpenAI, Anthropic, Vertex AI, and university/NSF ACCESS HPC.

CADS: Context-Aware Detection of Specialized Language	Summer 2026 – Present
Ongoing REU research mentorship with Valerie Lopez, University of Central Florida	

- Developed a pipeline combining translation instability, surface clues, masked-language-model likelihood, and model agreement, with logistic-regression fusion.
- Reported 80.1 F1 on GenZ-A; broader domain and multilingual tests exposed limits in transfer.

BIAS-101: Demographic Bias in Video Action Recognition	Jan 2025 – Apr 2025
Processed 85,000+ candidate videos and added 2,000+ curated clips; used multimodal annotation and statistical audits to study UCF-101 representation.	

DETECT: Real-Time Gaze and Facial-Signal Analysis	Aug 2024 – Apr 2025
Led a 6-member team; built a MediaPipe FaceMesh pipeline and deployed an Astro, Go, PostgreSQL, and Docker application. Project evaluation reported 80%+ gaze-tracking accuracy.	

Technical Skills

AI and evaluation: Social reasoning, Theory of Mind, LLM/VLM evaluation, benchmark design and QA
NLP: Machine translation, text classification, masked-LM scoring, named entity recognition, fuzzy matching, model fusion
Programming: Python, C/C++, Java, JavaScript, Go, SQL
Platforms: Azure OpenAI, Anthropic API, Google Vertex AI, Hugging Face, Azure, Google Cloud, HPC
Tools: Git/GitHub, Linux, Docker, PostgreSQL, REST APIs, LaTeX, Jira